SUPPOSITIONS

STANDARD PROOF TECHNIQUES IN DISCRETE MATHEMATICS AND EXAMPLE APPLICATIONS

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ABSTRACT

This document examines standard proof techniques found in discrete mathematics. It combines meticulous explanations with example applications to accessibly define and demonstrate each approach.

CONTENTS

Proof Techniques for Existential Statements 1
Proof by Construction 2

Proof Techniques for Universal Statements 3
Proof by Exhaustion 4
Direct Proof 5
Proof by Division into Cases 6
Proof by Contraposition 7
Proof by Contradiction 8
Proof by Mathematical Induction 9
Proof of Subset Containment 10
Proof of Injectivity 11
Proof of Surjectivity 12
Proof of Bijectivity 13

Example Problems 14
Example Problem 1: To-Be Determined 15
Example Problem 2: To-Be Determined 16
Example Problem 3: To-Be Determined 17
Example Problem 4: To-Be Determined 18
Example Problem 5: To-Be Determined 19
Example Problem 6: To-Be Determined 20

PROOF TECHNIQUES FOR EXISTENTIAL STATEMENTS

PROOF BY CONSTRUCTION

PROOF TECHNIQUES FOR UNIVERSAL STATEMENTS

PROOF BY EXHAUSTION

DIRECT PROOF

PROOF BY DIVISION INTO CASES

PROOF BY CONTRAPOSITION

PROOF BY CONTRADICTION

PROOF BY MATHEMATICAL INDUCTION

PROOF OF SUBSET CONTAINMENT

PROOF OF INJECTIVITY

PROOF OF SURJECTIVITY

PROOF OF BIJECTIVITY

EXAMPLE PROBLEMS

EXAMPLE PROBLEM 1: TO-BE DETERMINED

Exercise. Prove that if a is an odd integer, then $a^2 = 8m + 1$ for some $m \in \mathbb{Z}$.

Proof. Hello! □

EXAMPLE PROBLEM 2: TO-BE DETERMINED

Exercise. Prove that if $x < -4$ and $y > 2$, then the distance from (x, y) to $(1, -2)$ is at least 6.

Proof. Hello! □

EXAMPLE PROBLEM 3: TO-BE DETERMINED

Exercise. A point $(m, n) \in \mathbb{R}^2$ is a lattice point if $m, n \in \mathbb{Z}$. Prove that the number of lattice points inside any circle centered at the origin equals $4k + 1$ for some $k \in \mathbb{Z}$.

Proof. Hello! □

EXAMPLE PROBLEM 4: TO-BE DETERMINED

Exercise. For $(a_1, a_2, \ldots, a_n), (b_1, b_2, \ldots, b_n) \in \mathbb{R}^n$, the dot product is

$$(a_1, a_2, \ldots, a_n) \cdot (b_1, b_2, \ldots, b_n) = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n.$$

For example, $(5, 2, -3) \cdot (3, 7, 4) = 5 \cdot 3 + 2 \cdot 7 + (-3) \cdot 4 = 17$. Prove that

$$(1, 2, 3, \ldots, n) \cdot (1!, 2!, 3!, \ldots, n!) = (n+1)! - 1$$

for all positive integers n.

Proof. Hello! □

EXAMPLE PROBLEM 5: TO-BE DETERMINED

Exercise. Let A be a nonempty set. Prove that a function $f\colon A \to B$ is injective if and only if f has a left inverse, that is, a function $g\colon B \to A$ such that $(g \circ f)(x) = x$ for all $x \in A$.

Proof. Hello! □

EXAMPLE PROBLEM 6: TO-BE DETERMINED

Exercise. A set $S \subseteq \mathbb{R}$ is open if for each $b \in S$, there exists $\epsilon > 0$ such that $(b-\epsilon, b+\epsilon) \subseteq S$. Prove that the intersection of two open sets is open.

Proof. Hello! □